Notes on *Dynamic Immunization Node Model for Complex Networks Based on Community Structure and Threshold*

1 Motivation

1.1 Network Immunization Problem

Definition. Given a network represented by an undirected graph $G = (V, E)$, the *network immunization problem* aims to select an *immunized set* $V_P \subseteq V$ that minimizes the final number of infected nodes V_I.

Objective. Find

$$\min_{V_p \subseteq V} \mathbb{E}[|V_I|]$$

by dynamically determining how many nodes to immunize using a threshold.

1.2 Existing Strategies and Hypothesis

Traditional immunization strategies include:

- **Static Immunization (SI).** Select k nodes to immunize before propagation begins.
- **Uniform Immunization (UI).** Given the total propagation time T, immunize $\frac{k}{T}$ nodes at each time step.
- **Exponent Immunization (EI).** To prioritize earlier stages of propagation, immunize $k \cdot 2^{-t}$ nodes at time step t.

However, these methods either (1) select nodes in advance, or (2) predetermine the number of immunizations at each time step. They cannot fully adapt to changeing infection states. Furthermore, traditional propagation models often assign transmission probabilities without sufficiently considering the relationships between connected nodes.

Therefore, Shang et al. hypothesize that combining *community structure*, *node similarity*, and *dynamic immunization gain* can improve the effectiveness and efficiency of network immunization.

2 Methodology

2.1 Community Structure and Node Similarity

Definition. A *community* is a group of nodes with dense internal connections and relatively sparse external connections.

- Shang et al. assumes that harmful information is more likely to spread between nodes belonging to the same community.

Similarity Measure. Shang et al. uses *cosine similarity* to measure the structural similarity between nodes u and v:

$$\mathrm{Sim}(u, v) = \frac{|N(u) \cap N(v)|}{\sqrt{|N(u)| \cdot |N(v)|}}$$

- $N(u), N(v)$ are the neighbor sets of nodes u and v, respectively.
- $|N(u) \cap N(v)|$ is the number of common neighbors between nodes u and v.

2.2 Community Detection

Definition. *Community detection* is the process of dividing a network into communities.

Definition. *Modularity density* is a measure of how well a network is divided into communities:

$$D = \sum_{c \in C} \frac{2m_c - e_c}{n_c}$$

- C is the set of all communities.
- m_c is the number of edges within community c.
- e_c is the number of edges connecting community c to other communities.
- n_c is the number of nodes in community c.

Algorithm. To detect communities, Shang et al. uses a *community integration algorithm*:

1. **Initialize communities.** Identify core nodes based on their degree and group nodes using their similarity.
2. **Evaluate neighboring communities.** Calculate the change in modularity density, ΔD, that would result from merging two communities.
3. **Merge communities.** Combine neighboring communities if $\Delta D \geq 0$.
4. **Repeat.** Continue this merging process until no more merges are possible.

2.3 Community-Based Propagation Model

Definition. An *community-based propagation model* is a propagation model in which transmission probabilities depend on node similarity and whether two nodes belong to the same community. For nodes u and v:

$$p_{uv} = \begin{cases} \frac{\text{Sim}(u,v)}{2} + \frac{1}{|N_u|} & \text{if } j \in N_u \\ \frac{\text{Sim}(u,v)}{2} & \text{otherwise} \end{cases},$$

where N_u is the neighbors of node u that belong to the same community. Through this model, nodes in the same community have a higher transmission probability.

Definition. Given a set of infected neighbors $N_{\text{in}}(u)$, the *activation probability* is the probability that node u becomes infected:

$$AP(u) = 1 - \prod_{v \in N_{\text{in}}} (1 - p_{uv})$$

Node u becomes *infected* when its activation probability exceeds the propagation threshold pt:

$$AP(u) > pt.$$

2.4 Modified Immunization Gain

Definition. *Modified Immunization Gain* (MIG) measures how much immunizing a node would reduce the infection probabilities of its uninfected neighbors.

Definition. The *influence* of node u on its neighbor v is

$$Ia(u,v) = \prod_{w \in N(v).w \neq u} (1 - AP(w)) \cdot p_{vw} - \prod_{w \in N(v)} (1 - AP(w)) \cdot p_{vw},$$

i.e. the difference in node v's infection probability with and without node u's contribution.

Definition. The *total MIG* for a node u is

$$MIG(u) = \sum_{v \in N_{\text{out}}(u)} Ia(u,v),$$

where $N_{\text{out}}(u)$ is the set of node u's neighbors that are not infected. A high MIG means that immunizing u is expected to substantially reduce infection risk among its neighbors, while a low MIG means that immunizing u provides relatively small immediate benefit.

2.5 Dynamic Threshold-Based Immunization

Definiton. *Dynamic immunization based on Community structure and Threshold* (NICT) immunizes a candidate node u when its modified immunization gain exceeds a specified immune thresohld it, i.e.

$$MIG(u) > it.$$

Algorithm. (Dynamic immunization heuristic)

1. Detect community structure and calculate node similarities using the community integration algorithm.
2. Compute the propagation probabilities between connected nodes using the above definition.
3. Generate the initial infected node set.
4. At each time step t, calculate the MIG for the uninfected neighbors of currently infected nodes.
5. Immunize nodes whose MIG exceeds the immune threshold.
6. Continue propagation and update the infected node set.
7. Repeat this process until the maximum propagation time is reached.

3 Experimental Design

3.1 Propagation Model

Definition. Shang et al. uses an *intrusion-node propagation model* with the following components:

- *Intrusion nodes* are external nodes that introduce harmful information into the network.
- The initial infected nodes are existing nodes in the network that are infected by the intrusion nodes.
- Infected nodes propagate according to the community-based propagation probabilities defined above.
- After a node is immunized, it cannot become infected nor transmit harmful information to other nodes.

Procedure. For each experiment,

1. Detect communities and calculate the node similarities.
2. Calculate the propagation probabilities between connected nodes.
3. Randomly select the initial infected nodes based on the intrusion model.
4. At each time step t, immunize eligible neighbors of infected nodes using MIG and the defined immune threshold.
5. Simulate propagation and update the infected node set
6. Record the final number of infected and immunized nodes.
7. Average the results over 20 runs.

3.2 Datasets and Baselines

Networks. Shang et al. evaluate NICT on eight empirical networks (Karate, Dolphin, Football, SFI, Netscience, Power, PGP, and the Internet), where the network sizes range from 34 to 22,963 nodes.

Baselines. They compare NICT against five immunization algorithms: SI, UI, EI, Dynamic Immunization (DI), and Formal Position-Based Acquaintance-Degree (FPAD).

Parameters. (Propagation threshold, immune threshold, and propagation steps)

- $pt = 0.5$, selected after examining propagation under different threshold values.
- it was selected according to each network's average node degree.
- $T = 10, 50, 100$ for small, medium, and large-scale networks, respectively.

Evaluation Metric. Shang et al. uses the *Node Immune Effect* (NIE), which accounts for both the final number of infected nodes and the number of immunized nodes:

$$NIE = \sqrt{\frac{n}{I}} + \sqrt{\frac{n}{IM}}.$$

- n is the total number of nodes in the network.
- I is the final number of infected nodes.
- IM is the final number of immunized nodes.

4 Results and Limitations

4.1 Experimental Findings

Propagation Model Comparison. (Figures 6-7)
Shang et al. compare their community-based propagation model against the Weighted Independent Cascade (WIC) model on the Karate network:

- WIC infects 32 of the 34 nodes.
- The community-based propagation model activates $\sim 71\%$ of nodes.

Shang et al. argues that incorporating community structure and similarity produces a more realistic distribution of infections over time.

Dynamic Immunization Performance. (Figures 8-13)
On the Power network, NICT immunizes 10-23 nodes across 20 runs, with 50-83 final infected nodes.

- SI and FPAD exhibit greater variation in infection outcomes than NICT.
- EI achieves relatively stable results, demonstrating that other dynamic scheduling strategies can also perform well.

Overall Results. (Tables VI-VII)

- NICT achieves the highest best-case NIE across all eight networks.
- NICT achieves the highest average NIE in five of the eight networks.

Shang et al. conclude that dynamic immunization based on community structure and MIG can improve intervention efficiency compared with static immunization strategies.

4.2 Limitations

1. MIG measures the immediate reduction in infection probability among neighboring nodes, so a node with modest local influence but substantial downstream influence may be undervalued.

2. The propagation model assumes that information is more likely to spread within communities and between structurally similar nodes. However, actual transmission probabilities may depend on additional factors (e.g. individual activity, temporal interactions).

3. *pt* and *it* significantly influence the results; in particular, *it* must be selected according to the network characters, and the authors do not establish a universally optimal threshold.

4. NICT determines the number of immunized nodes dynamically using a threshold, rather than optimizing under a fixed budget. This makes it difficult to guarantee that the method will follow practical resource constraints in real world settings.

5. The NIE metric becomes undefined when the number of infected or immunized nodes is zero and can produce misleadingly high values when very few nodes are immunized.